Coherent Multi-Horizon Forecasting of Power Outage Severity with State-Wise Unbalanced Optimal Transport

Team SIOL

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INFORMS 2026 Data Mining Society Data Challenge Finalist Presentation

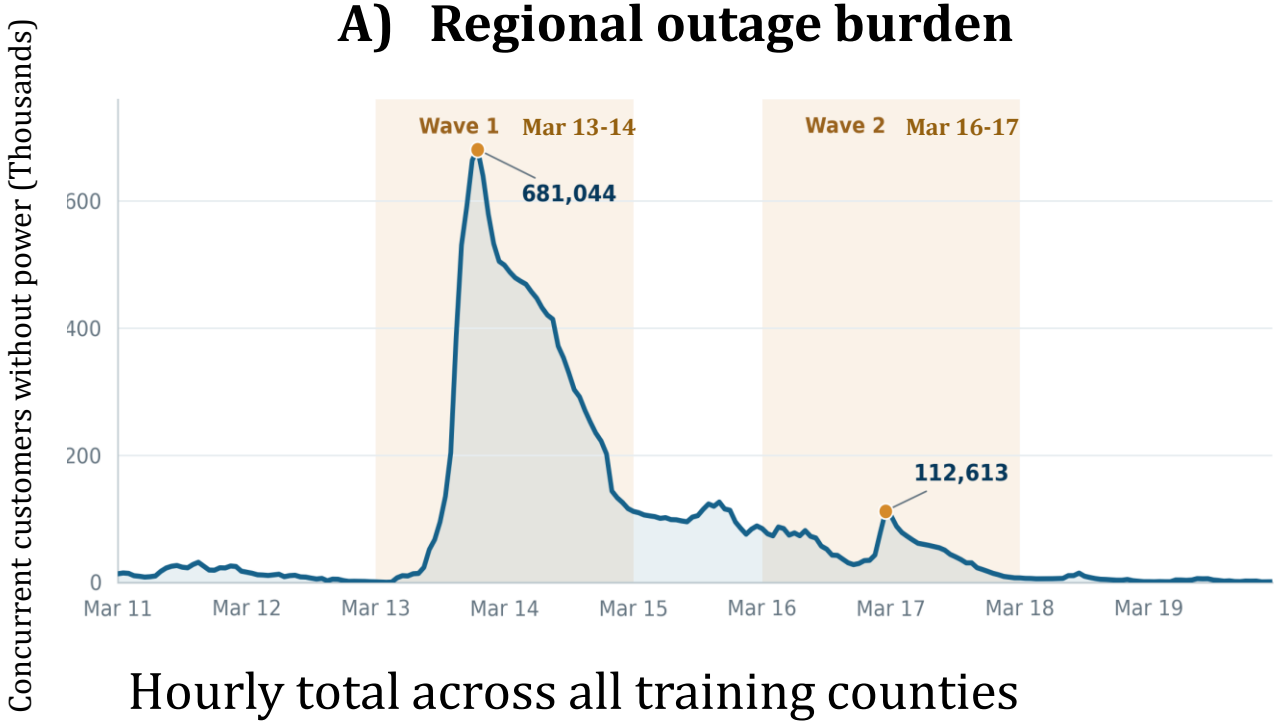
October 31, 2026

Motivation

Outage Dynamics: Abrupt Temporal Changes and Geographic Heterogeneity

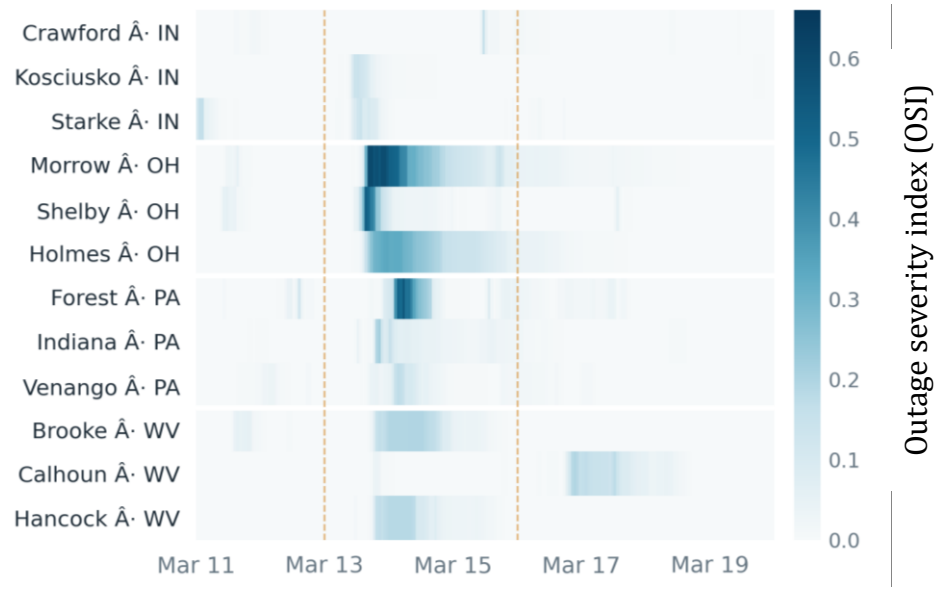
Observation: outage activity surges abruptly, while severity follows different trajectories across counties

A) Regional outage burden



Hourly total across all training counties

B) County-level severity



Three highest event-peak OSI counties per state

Forecasts must capture rapid temporal escalation and spatially uneven severity across counties

Problem Formulation

County-Held-Out Multi-Horizon OSI Forecasting

A) Dataset

State	Train	Test
IN	72	20
OH	70	18
PA	52	15
WV	45	10
TOTAL	**239**	**63**

B) Information available for each test county

	March 11-13	March 14-19
	72 h HISTORY	**144 h Forecast ORIGINS**
Weather	Observed	Available
Outage state	Observed	Withheld
OSI forecast	—	Predict OSI at four horizons

C) Forecast targets for each test county and each forecast origin t (Mar 14-19)

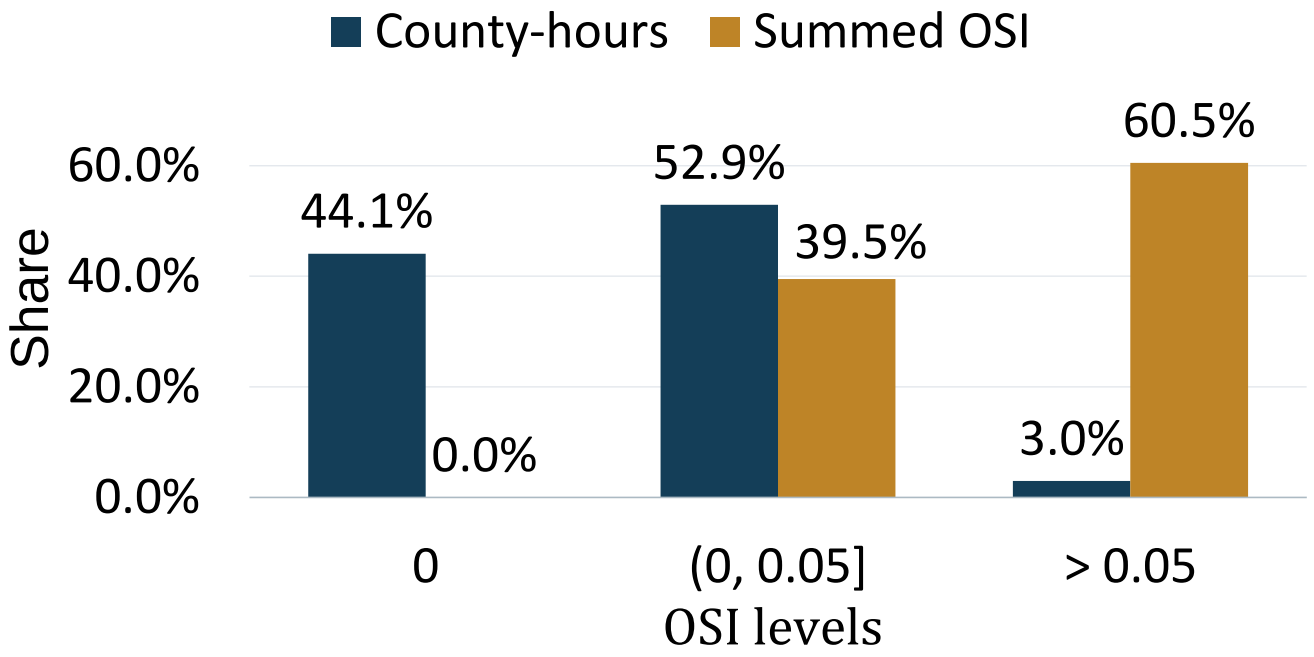
t + 1h	t + 6h	t + 24h	t + 48h
9009 scored	8694 scored	7560 scored	6048 scored

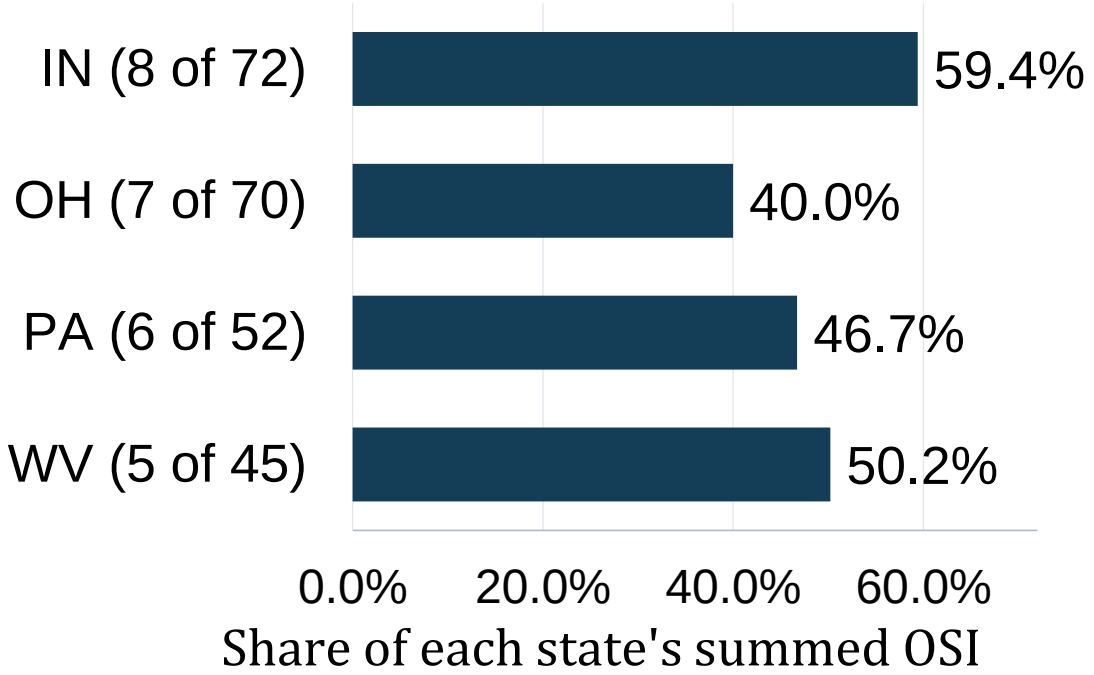
9,072 submission rows (144 origins × 63 counties). Targets after March 19, 23:00 remain NaN.

Data Characteristics

Rare Severe Hours and Uneven County Burden

Observation: Severity is concentrated in a small fraction of observations and counties





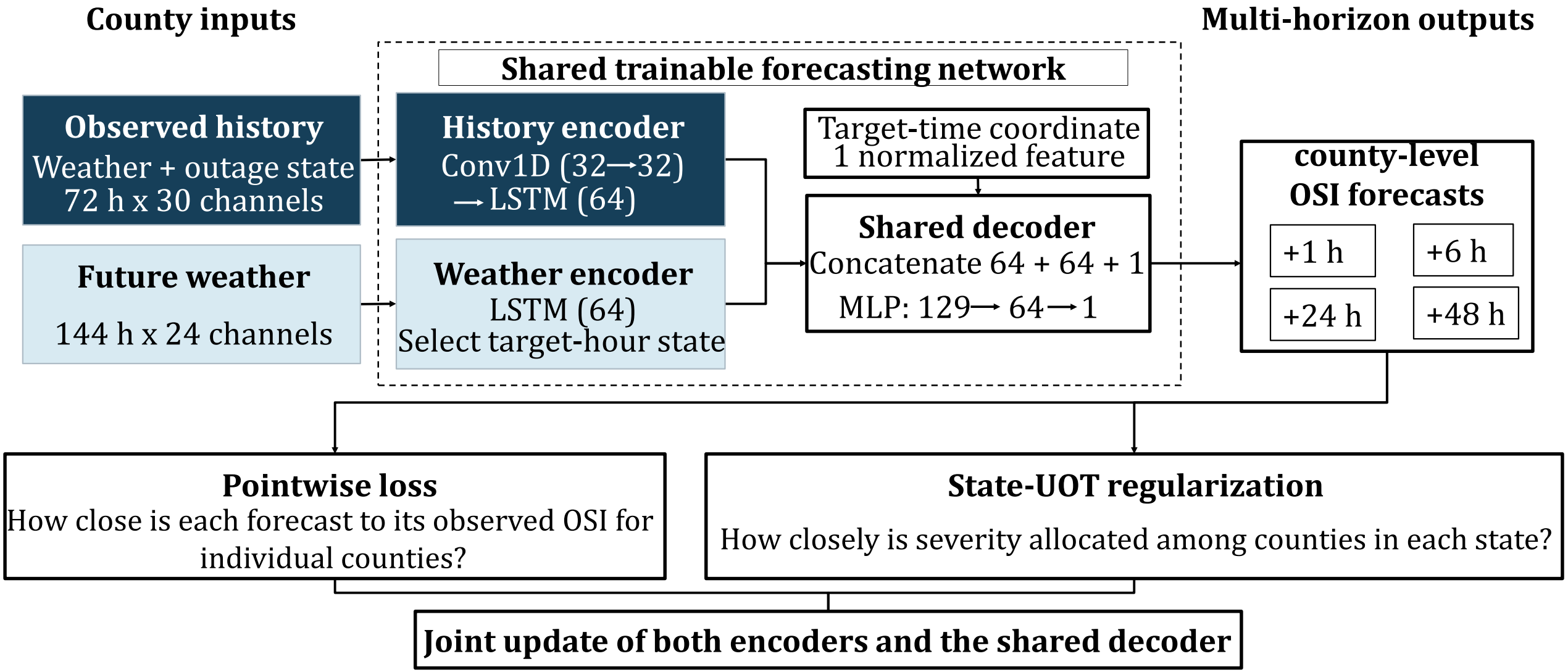
3.0% of county-hours carry *60.5%* of summed OSI

Within each state about *10%* of counties carry *40-59%*

A forecasting model must accurately predict severity at individual counties, while also representing how severity is distributed geographically.

Methodology

End-to-End Dual-Stream Forecasting Framework with State-UOT



Methodology

Dual-Stream Forecasting with Target-Time Consistency

History Encoder (Conv1D + LSTM)

$$u_c = E_{\theta_H}(H_c) \in \mathbb{R}^{64}$$

Weather Encoder (LSTM)

$$v_{c,\tau} = [G_{\theta_W}(W_c)]_\tau \in \mathbb{R}^{64}$$

c: county, t: origin, h: horizon, θ: model parameters,
H: observed history, W: supplied future-weather

Shared Prediction Function

$$\hat{z}_{c,t,h} = D_{\theta_D}([\, u_c \,;\, v_{c,\tau} \,;\, \xi(\tau) \,]), \qquad \tau = t + h$$

Where: $\xi(\tau) = \frac{\tau}{143}$, $\quad 0 \le \tau \le 143$

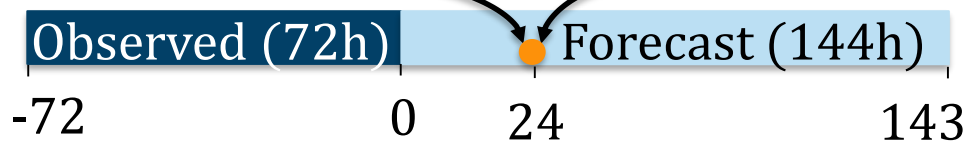
$$\hat{y}_{c,t,h} = \max(0, e^{\hat{z}_{c,t,h}} - 1)$$

Where: $\hat{z}$: prediction of $\log(1 + \text{OSI})$, $\quad \hat{y}$: raw OSI

Same physical target, consistent forecast

Request 1	
origin	t=0
horizon	+24
target	$\tau = 24$

Request 2	
origin	t=18
horizon	+6
target	$\tau = 24$



Both requests use the same inputs

u_c (county history)	$v_{c,24}$ (target-hour Weather)	$\xi(\tau) = 24/143$ (normalized time)

$$t + h = t' + h' \implies \hat{y}_{c,t,h} = \hat{y}_{c,t',h'}$$

Methodology

County Accuracy and State Severity: Two Learning Signals

1. County accuracy

$$L_{pointwise\ loss} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \ell_{SL1}(\hat{z}_i - log(1 + y_i))$$

$\mathcal{V}$: valid county–origin–horizon requests, $\hat{z}$: log prediction, y: observed OSI

2. State severity comparison

$$OT_{\varepsilon,\rho}(a,b) = \min_{\pi \geq 0} \{\langle C, \pi \rangle + \varepsilon\, \mathrm{KL}(\pi \parallel ab^{\top}) + \rho\, [\mathrm{KL}(\pi 1 \parallel a) + \mathrm{KL}(\pi^{\top} 1 \parallel b)]\}$$

$$S_{\varepsilon,\rho}(a,b) = OT_{\varepsilon,\rho}(a,b) - \frac{1}{2}[OT_{\varepsilon,\rho}(a,a) + OT_{\varepsilon,\rho}(b,b)] + \frac{\varepsilon}{2}(m(a) - m(b))^2$$

Predicted raw OSI: $a_c = \hat{y}_{c,\tau} + \eta$, Observed raw OSI: $b_c = y_{c,\tau} + \eta$, $\eta = 10^{-8}$

Methodology

Joint End-to-End Optimization

Joint optimization of the forecasting network

$$L_e(\theta) = L_{pointwise}(\theta) + \lambda_e L_{State-UOT}(\theta)$$

Where State–UOT loss is:

$$L_{State-UOT} = \sum_{s:\, \mathcal{B}_s \neq \emptyset} \frac{|\mathcal{B}_s|}{|\mathcal{B}|} \frac{1}{|T|} \sum_{\tau \in T} S_{\varepsilon,\rho}(a_{s,\tau}, b_{s,\tau})$$

$\mathcal{B}$: minibatch of counties $\mathcal{B}_s$: counties from state s T: set of 143 unique future target hours

Validation Design

County Validation and Final Prediction

County-level validation

239 labeled training counties

	Group 1	Group 2	Group 3	Group 4	Group 5
Fold 1	VALID	TRAIN	TRAIN	TRAIN	TRAIN
Fold 2	TRAIN	VALID	TRAIN	TRAIN	TRAIN
Fold 3	TRAIN	TRAIN	VALID	TRAIN	TRAIN
Fold 4	TRAIN	TRAIN	TRAIN	VALID	TRAIN
Fold 5	TRAIN	TRAIN	TRAIN	TRAIN	VALID

Each county validates once

↓

Held-out predictions

RMSE and MAE at four horizons

Final training and submission

All 239

training counties

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5 trained models

63 holdout counties

72 h observed history
144 h future weather

Future outages hidden

Average 5 model forecasts

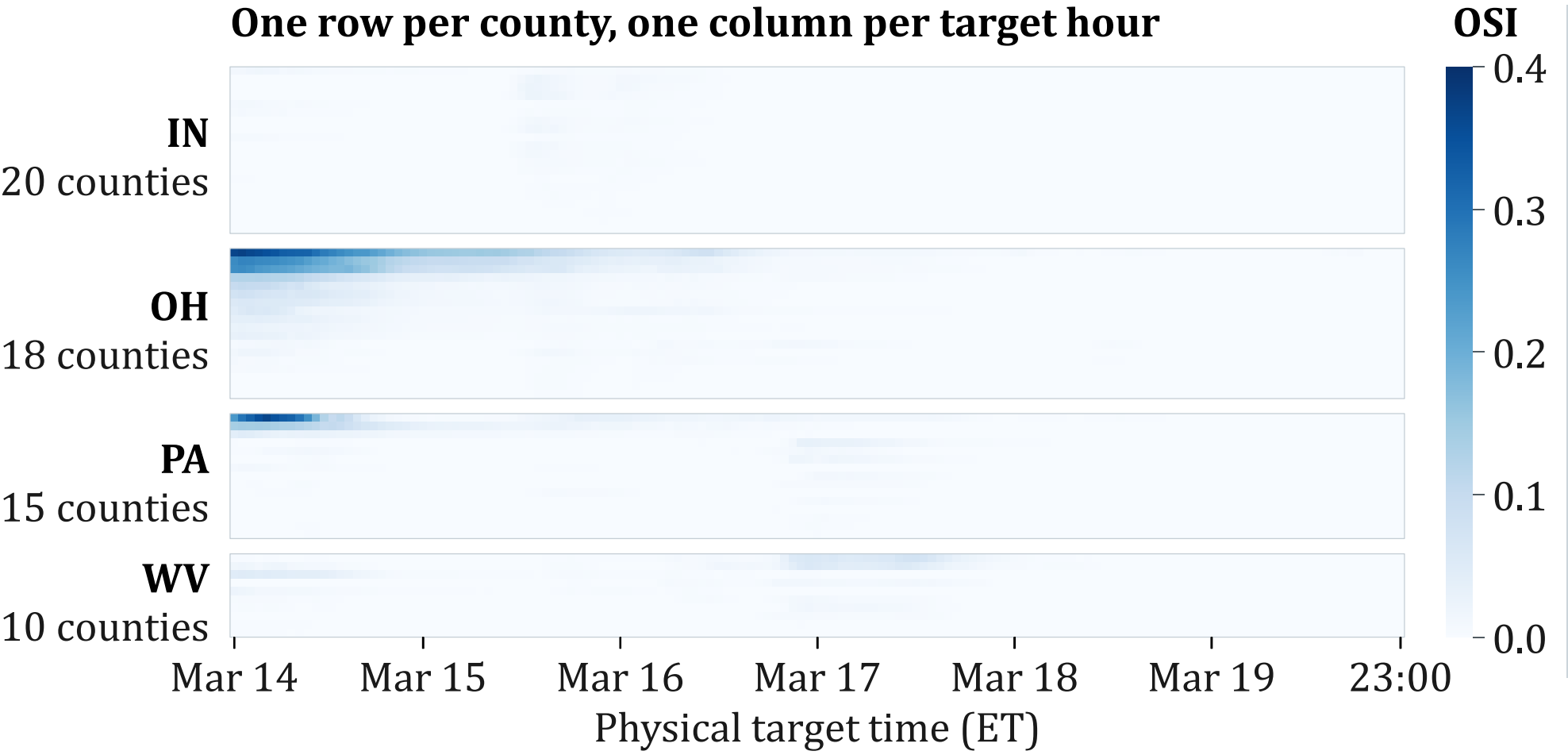
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Final submission

9,072 rows at +1, +6, +24, +48 h

Holdout Forecasts

Forecast Severity Across 63 Unseen Counties



Forecast summary

Share of summed OSI

67.4%

Top 7 of 63 counties

59.9%

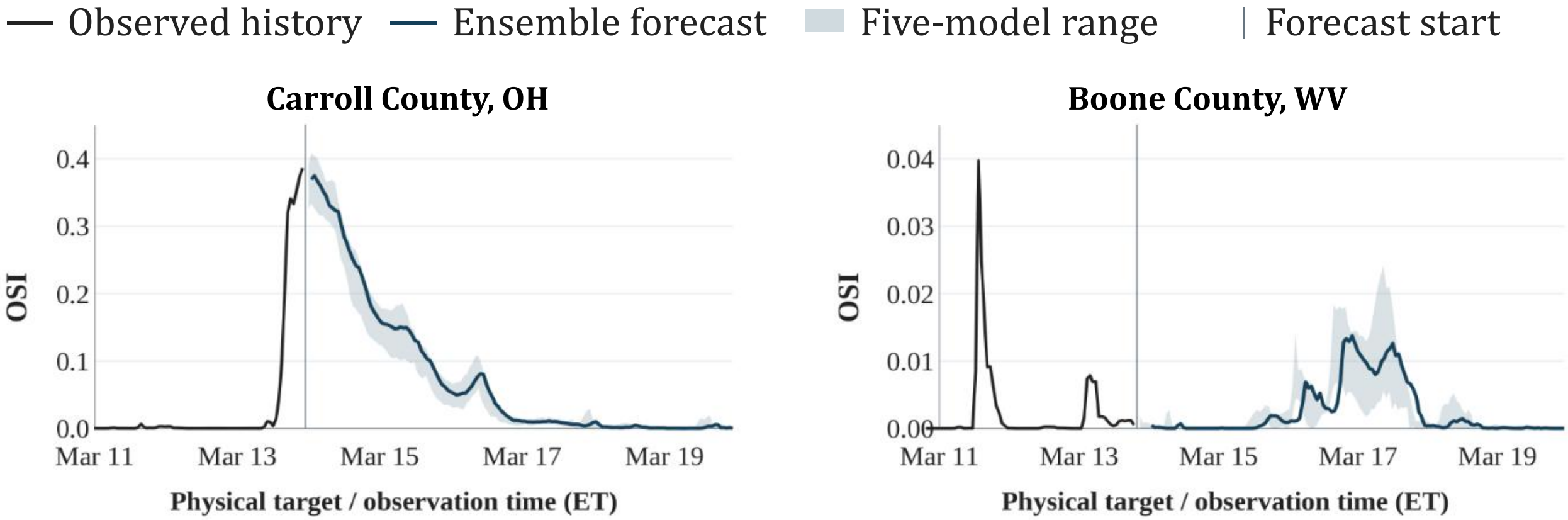
First 24 target hours

Submission checks

Negative forecasts	0
Same-target gap	0

Holdout Forecasts

Two Contrasting County Forecast Trajectories



High initial severity, then predicted decline

Low initial severity, later predicted escalation

Conclusion

One Joint Model for Coherent County Outage Forecasts

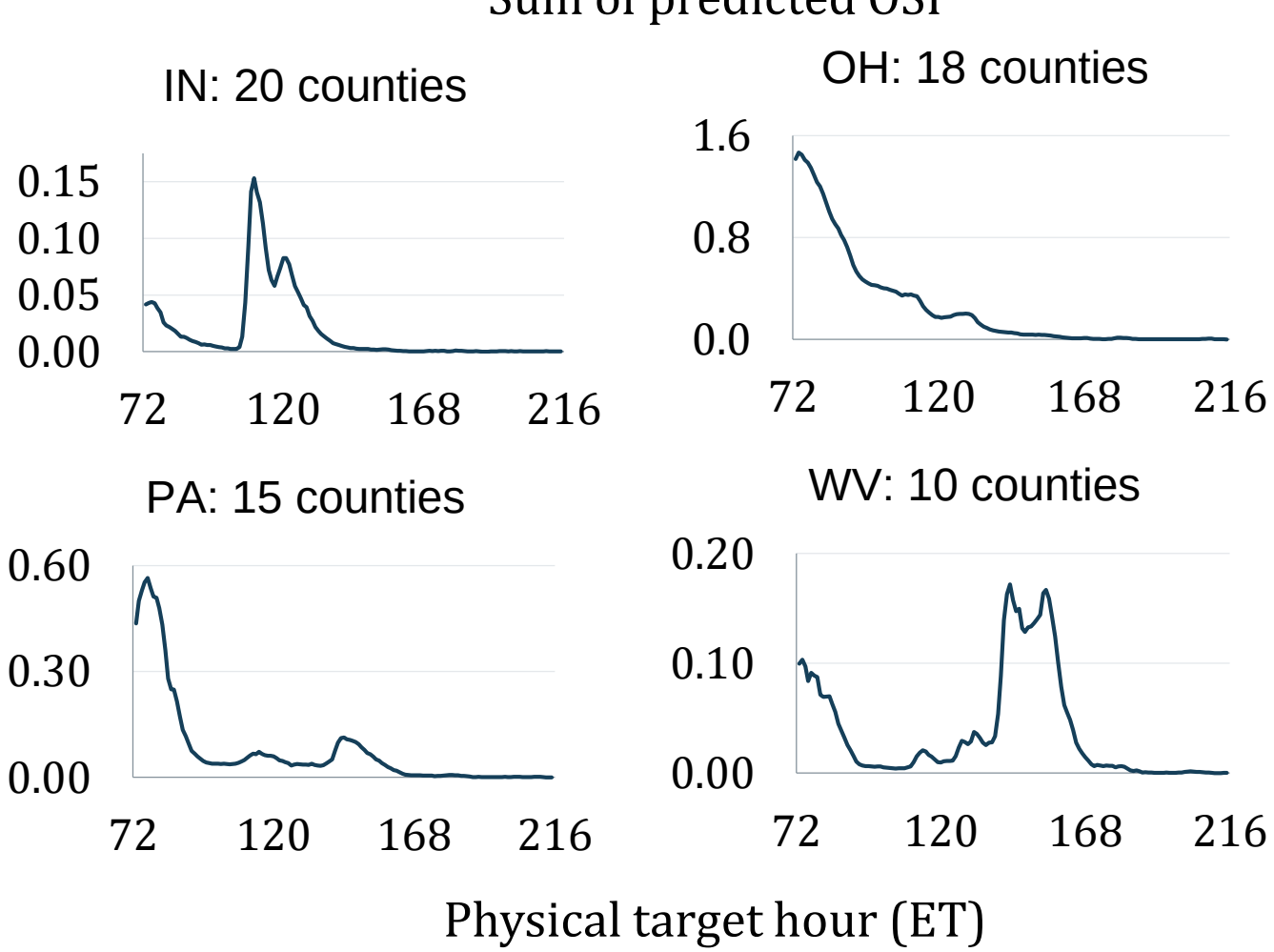
We developed an end-to-end dual-stream forecasting framework that

- Preserves consistency across forecast horizons by design.
- Aligns predicted and observed severity across counties within each state using unbalanced optimal transport.

Future direction:

Evaluate on additional storms and connect forecasts to resource allocation.

Final forecasts aggregated by state



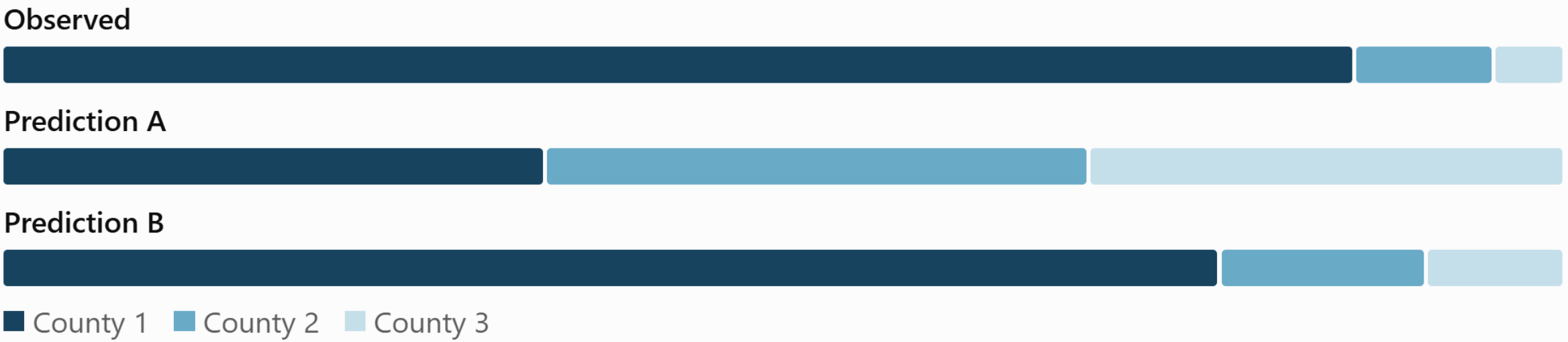
Thank you!

I welcome your questions
and opportunities to collaborate.

Research and contact

Objective 2 — State-conditioned UOT

County	Observed	Prediction A	Prediction B
1	.20	.08	.18
2	.02	.08	.03
3	.01	.07	.02
Total	**0.23**	**0.23**	**0.23**



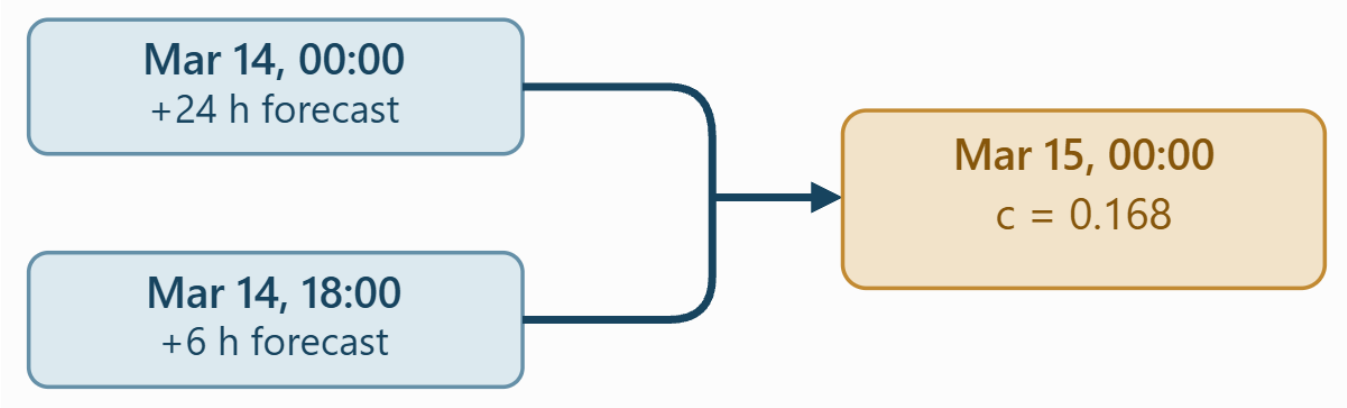
State-UOT term compares predicted and observed severity masses within each state during training.

Target-time Coordinate

What the decoder knows when it makes a prediction



Different forecast origins and horizons, but the same county and physical target time.



The model selects the same future-weather LSTM state for that target hour and uses the same history representation → Decoder produces the same prediction for both

Imagine a training minibatch containing 10 counties.

Illustrative minibatch composition

Ohio: 6	Indiana: 4

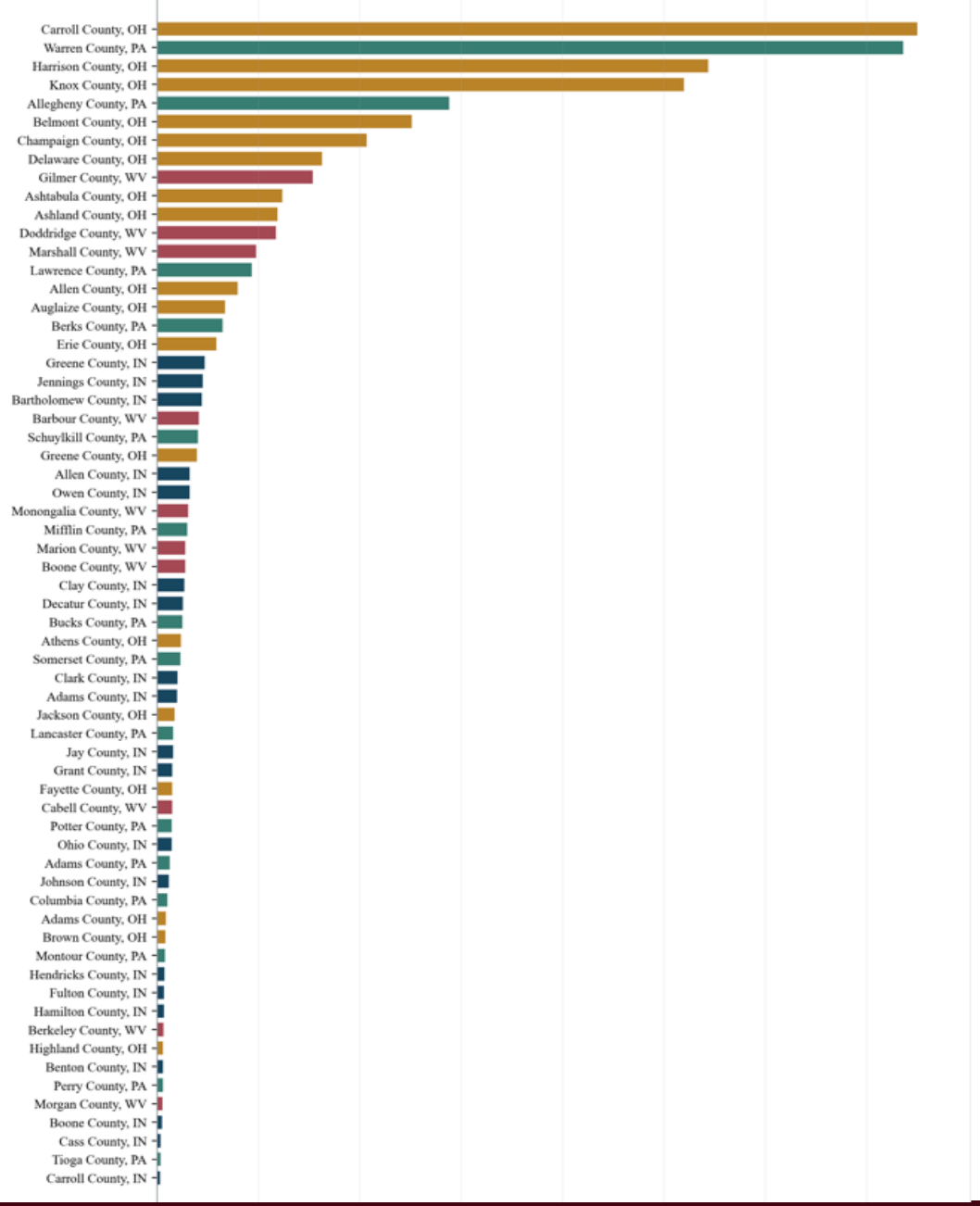
For example, suppose our minibatch includes three counties from Ohio at a particular target hour.

Illustrative state-level severity comparison

County	Predicted OSI	Observed OSI
County A	0.15	0.20
County B	0.08	0.02
County C	0.05	0.01

$$a_{s,\tau} = \begin{bmatrix} 0.15 \\ 0.08 \\ 0.05 \end{bmatrix}, \qquad b_{s,\tau} = \begin{bmatrix} 0.20 \\ 0.02 \\ 0.01 \end{bmatrix}$$

$$L_{\text{State-UOT}} = 0.6\,\overline{S}_{\text{Ohio}} + 0.4\,\overline{S}_{\text{Indiana}}$$


Carroll County, OH
Warren County, PA
Harrison County, OH
Knox County, OH
Allegheny County, PA
Belmont County, OH
Champaign County, OH
Delaware County, OH
Gilmer County, WV
Ashtabula County, OH
Ashland County, OH
Doddridge County, WV
Marshall County, WV
Lawrence County, PA
Allen County, OH
Auglaize County, OH
Berks County, PA
Erie County, OH
Greene County, IN
Jennings County, IN
Bartholomew County, IN
Barbour County, WV
Schuylkill County, PA
Greene County, OH
Allen County, IN
Owen County, IN
Monongalia County, WV
Mifflin County, PA
Marion County, WV
Boone County, WV
Clay County, IN
Decatur County, IN
Bucks County, PA
Athens County, OH
Somerset County, PA
Clark County, IN
Adams County, IN
Jackson County, OH
Lancaster County, PA
Jay County, IN
Grant County, IN
Fayette County, OH
Cabell County, WV
Potter County, PA
Ohio County, IN
Adams County, PA
Johnson County, IN
Columbia County, PA
Adams County, OH
Brown County, OH
Montour County, PA
Hendricks County, IN
Fulton County, IN
Hamilton County, IN
Berkeley County, WV
Highland County, OH
Benton County, IN
Perry County, PA
Morgan County, WV
Boone County, IN
Cass County, IN
Tioga County, PA
Carroll County, IN